Transport Restructuring and Regional Development: Evidence from Argentina, 1960–1991

(Working title; alternatives under discussion)

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Abstract

[PLACEHOLDER: Abstract – write last. (1) setting, (2) method, (3) MA elasticity, (4) counterfactual rail vs road, (5) local infrastructure mechanisms. Under 150 words.**]**

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1 Introduction

How do large-scale transport infrastructure changes reshape regional economies? While most research examines infrastructure expansions designed to promote development, governments often face a different challenge: fiscal consolidation that forces difficult choices about which infrastructure to maintain. Argentina's experience between 1960 and 1991 provides a unique window into this question. Facing severe fiscal pressure, the government implemented a massive transport restructuring: closing [10,000] kilometers of railway lines while expanding paved roads by [18,000] kilometers. This episode offers rare evidence on how fiscal consolidation-driven infrastructure changes—rather than development-oriented investments—affect regional market access and economic outcomes.

This setting is valuable for two reasons. First, it speaks to a policy-relevant question: when governments must cut infrastructure spending, how do these cuts reshape the economic geography? The Argentine plan, motivated by fiscal rationalization rather than regional development goals, effectively acted as an unintended industrial and regional policy by shifting the dominant transport mode from rail to road. Second, it provides an opportunity to estimate how regional outcomes respond to market access changes in a multimodal infrastructure shock. Unlike typical infrastructure studies that examine single-mode expansions, Argentina's bundled rail-road restructuring allows us to study elasticities of population and sectoral employment to market access changes that integrate both transport modes.

Estimating the causal effects of such infrastructure changes faces two fundamental challenges. First, infrastructure placement and removal decisions are endogenous. Even when motivated by fiscal consolidation, governments may close lines in already-declining regions or expand roads where growth is expected. This makes it difficult to separate the causal effect of market access changes from pre-existing economic trends.

Second, when both rail and road networks change simultaneously, their effects are difficult to disentangle. The changes are often spatially correlated—regions losing rail access may or may not gain road access—creating multicollinearity problems if we try to estimate mode-specific effects in a single regression. Moreover, the theoretical framework suggests that what matters for regional outcomes is total market access integrating all transport modes, not the separate contribution of each mode. This raises a methodological question: what can and cannot be identified about mode-specific effects in multimodal infrastructure episodes?

This paper addresses these challenges by studying Argentina's transport restructuring through the lens of market access. We construct a unified market access index that integrates both rail and road networks using a standard spatial economics framework, computing generalized transport costs and shortest-path routes on the historical networks. Our main analysis estimates how regional outcomes—population, sectoral employment, and economic activity—respond to changes in total market access induced by the observed infrastructure plan. We then use counterfactual market access measures to explore the relative roles of rail closures versus road expansion, constructing separate scenarios where only rail changes (roads frozen) or only roads change (rail frozen).

Our empirical framework follows the market access approach standard in spatial economics and trade. We model each district's economic outcomes as functions of its market access, defined as the population-weighted sum of inverse generalized transport costs to all other districts. To address endogeneity, we instrument for observed market access changes using two sources of variation. First, we exploit the Larkin Plan's technical rules: the World Bank study that motivated the reforms only examined railway segments below a specific freight threshold for potential closure, creating a discontinuity in closure probability. Second, we construct hypothetical road networks connecting major cities via least-cost paths, which predict road expansion but are orthogonal to local economic shocks.

We construct a novel dataset combining historical transport networks with census data at the district level, covering [312] districts across Argentina. We validate our identification strategy with pre-period balance checks, pre-trend tests, spatial placebo tests, and over-identification tests.

This paper contributes to several literatures. First, we provide causal evidence on the regional effects of infrastructure disinvestment driven by fiscal consolidation. The Argentine episode involved not just the removal of rail infrastructure but a technological shift in the dominant transport mode—from rail to road—that effectively acted as an unintentional industrial policy, reshaping the country's economic geography. Second, we contribute to understanding mode-specific effects in multimodal infrastructure episodes, proposing a counterfactual market access approach that clarifies what can and cannot be identified. Third, our findings speak to contemporary policy debates about modal shifts in transport, as many countries face ongoing transitions between rail and road. Fourth, we contribute to a longstanding debate in Argentine economic history about whether railroad closures caused the decline of hinterland towns or merely followed pre-existing trends.

The paper proceeds as follows. Section 2 provides historical context. Section 3 describes data and market access construction. Section 4 presents the empirical strategy. Section 5 reports main results. Section **??** explores mode-specific patterns. Section 7 examines mechanisms and heterogeneity. Section 8 concludes.

2 Historical Context

2.1 Argentina's Railway System and Fiscal Crisis

Argentina's railway network was one of the largest in the world. Construction began in 1857 and expanded rapidly through a combination of national capital, foreign investment—predominantly British and French—and direct government participation extending lines into rural areas that were not commercially profitable. By the early twentieth century, the network exceeded 40,000 kilometers, designed primarily to connect the agricultural hinterland of the Pampas, the Northwest, and the Northeast to the port of Buenos Aires. This radial, export-oriented design was both cause and consequence of the agricultural export boom of the first globalization: from 1869 to 1914, Argentine real exports grew approximately 500 percent (Fajgelbaum and Redding, 2018). By the start of the First World War, Argentina was the eighth richest country in the world.

Starting in the 1930s, export growth slowed considerably and profit margins of railroad companies declined. Investment in railroad infrastructure stalled, and the aging network experienced rising operational costs. Railroads also began to face competition from the automotive industry—Ford produced the first Model T in Latin America in Buenos Aires in 1925. In 1932, Law 11.658 created a National Trunk Road System and a lump-sum fuel tax whose proceeds were allocated entirely to road construction and improvement; by 1935, there were almost 3,000 kilometers of paved roads. During the Second World War, Argentina remained neutral and its young industrial sector flourished while factories in the developed world were retooled for war production. Railroads experienced a "second life" as wartime food shortages drove up international prices for Argentine agricultural exports, helping the country accumulate reserves of $1.6 billion (Whitaker, 2013). But by 1945, industry's contribution to GDP surpassed that of agriculture for the first time (Potash, 1969).

Supported by unions, the military, and the new industrial bourgeoisie, General Juan Domingo Perón reached the presidency in 1946. His economic policy rested on three pillars: strengthening the domestic market, centralized planning with state ownership of strategic sectors, and industrial development (Whitaker, 2013; Pahowka, 2005). The showcase of this program was the acquisition and nationalization of the private railroad companies in 1947 for $600 million—a purchase widely debated at the time, with economists including Raúl Prebisch arguing that the price was excessive given the crumbling state of the infrastructure (Rock, 1985). By the early 1950s, when the fortuitous advantages of the war years had faded, Argentina's economy slipped into depression. Perón moved to protect nascent industry, armoring the automotive and petrochemical sectors against international competition. The road network continued to grow at a slow, steady pace, reaching 9,000 kilometers of paved roads by 1955. The railroad network, meanwhile, remained stable at around 43,500 kilometers, but with a shrinking budget it suffered from poor maintenance and aging equipment.

In 1955, a coup removed Perón from government, and the country endured several years of political and economic instability. In 1958, Arturo Frondizi was elected president in an election from which Perón was banned. Frondizi's economic plan centered on fostering industrialization through foreign direct investment: Decree 3693 of 1959 promoted investment in the national automotive industry, and 23 new factories were approved within a year. His administration was equally aggressive in promoting road construction. In 1958, the lump-sum fuel tax was replaced by a 35 percent ad valorem tax (Decreto Nacional 505/1958), and in 1960 Law 15.274 instituted a sales tax on vehicles—in both cases, funds were permanently allocated to road construction and improvement. As for railroads, Frondizi initially sought to modernize the crumbling infrastructure, but the investment required to achieve a modern network while maintaining its current size was nearly prohibitive (López and Waddell, 2007). By 1960, annual railroad losses had reached approximately $280 million per year, accounting for almost 80 percent of the federal government deficit (Keeling, 1993).

2.2 The Larkin Plan (1960–1962)

In this context, the Argentine government agreed with the World Bank to commission a team of experts to evaluate the long-term economic and strategic viability of the national transport infrastructure. The team was led by General Thomas Larkin, an American expert in military logistics who had been decorated for his efforts supplying combat troops during the Second World War and had experience consulting on railroad modernization in France, Germany, and Japan. Between 1959 and 1961, the World Bank mission scrutinized the profitability of the existing railroad and road networks. The resulting study, published by the Public Works Ministry in 1961 as *Transportes Argentinos: Plan de Largo Alcance* ("Argentine Transportation: Long-Run Plan"), comprised three volumes totaling approximately 3,000 pages. The first volume contained an executive summary with diagnosis, methodology, results, and recommendations organized by transport mode and time horizon; the remaining two volumes, of approximately 1,000 pages each, provided detailed analysis.[1]

In a world of cheap and stable oil prices, and given the poor material and financial condition of the railroad system, the plan's main recommendations were dramatic: closure of approximately 15,000 kilometers of railroad track (roughly 32 percent of the existing network), dismissal of around 70,000 railroad employees (from a total of approximately 200,000), disposal of the entire fleet of steam vehicles, closure of most national railroad workshops and factories, and aggressive improvement and construction of paved roads—in some cases to substitute for closed railroad segments, but also to create new connections in the network (Larkin, 1962). The plan proposed the construction of approximately 10,000 kilometers of paved roads over ten years.

A key feature of the Larkin Plan's design is that it did not study the entire railroad network with equal intensity. Due to budget constraints and the vast size of the existing network, only railroad segments that fell below specific freight density thresholds were "studied" in detail—meaning that the experts checked each segment's explicit revenues and costs and analyzed whether closure was warranted. The thresholds were 1 million gross tons per year of cargo passing through a given segment, or 500 tons per kilometer per year of cargo entering the network through a given segment. Only 39.6 percent of the railroad network was studied. For each studied segment, the experts issued one of three recommendations: maintain, close, or perform a new study. Crucially, the likelihood of the plan recommending closure was much higher for studied segments than for nonstudied ones—a discontinuity that we exploit for identification.

The plan's recommendations were announced by President Frondizi in 1961, but massive protests and strikes erupted in many cities. After several months of clashes and social unrest, the government decided to abandon the plan. In the short term, only approximately 1,000 kilometers of track were closed. However, despite being abandoned after just a few months, the Larkin Plan was the only long-term comprehensive study of the national transport infrastructure for more than 30 years. It is therefore likely that its recommendations influenced national transport policy for decades ahead. As we document below, railroad closures in the 1960s and 1970s were approximately three times more likely for segments that had been studied by the plan, probably because no new large-scale study was ever undertaken to supersede it.

[1] We accessed physical copies of the plan at the Mariano Moreno National Library in Buenos Aires. We scanned and digitized two key objects: a map describing the operational railroad network in 1960 (our baseline) and three tables describing the diagnosis and recommendations for each railroad segment.

2.3 Implementation: Rail Closures and Road Expansion, 1960–1991

The Larkin Plan marked a trend break in the evolution of both transport networks (Figure **??**). For railroads, it initiated a long-term disinvestment and dismantling process. For roads, it started an almost two-decade period of aggressive construction and improvement. These large-scale changes occurred in a context of population growth (from 20 to 33 million between 1960 and 1991) and increasing GDP per capita (growing approximately 35 percent at an average annual rate of 0.95 percent).

Railroad closures proceeded in two distinct waves. The first wave, from 1960 to 1966, closed approximately 4,000 kilometers of track before political opposition brought the process to a halt. The second wave came under the military dictatorship that seized power in 1976: without union resistance, the military government ordered the immediate closure of over 6,000 additional kilometers of track between 1976 and 1979. After 1979, the network stabilized—no further closures or new construction occurred between 1979 and 1986. In total, the railroad network fell from approximately 43,500 kilometers in 1960 to approximately 33,500 kilometers by 1986, a reduction of roughly 23 percent.[2]

Road expansion, by contrast, was continuous throughout the period. Fueled by the dedicated fuel and vehicle taxes established under Frondizi, the paved road network grew from approximately 10,000 kilometers in 1954 to approximately 28,000 kilometers by 1986—an increase of roughly 180 percent. Gravel roads also expanded, though more modestly. By 1986, the combined paved and gravel road network had reached approximately 35,000 kilometers, roughly equal to the remaining railroad network. Importantly, the spatial pattern of road expansion did not simply mirror the pattern of railroad closures: roads were not systematically built to replace closed rail lines. As we show in Section 3, the correlation between rail closures and road expansion at the district level is weak, meaning that the two shocks affected largely different sets of districts.

2.4 Scale Economies and Sector-Mode Specialization: Motivation and Hypotheses

A feature of the Argentine setting that motivates part of our analysis is that railroads and roads may serve different economic sectors, owing to fundamental differences in their cost structures. Railroads exhibit strong scale economies: high fixed costs (track maintenance, stations, rolling stock) but low variable costs per ton-kilometer at high traffic density. Roads, by contrast, have more constant returns to scale—per-unit costs decline less steeply with volume. The Larkin Plan itself was designed around this logic: it studied segments precisely because their traffic density was too low for rail's scale economies to be realized.

Data from Baumgartner and Palazzo (1969), who estimated transport costs for Argentina using detailed operational data, are consistent with this pattern. At high cargo density (1,000 tons per

[2] We end our sample period in 1991 because there was a census in that year and because we wish to abstract from the potentially different effects of the privatizations of large fractions of the railroad and road networks during the 1990s.

day), railroad costs were 0.465 pesos per ton-kilometer compared to 1.000 for roads—rail was more than twice as cheap. At low cargo density (100 tons per day), the relationship reversed: railroad costs rose to 13.490 pesos per ton-kilometer while road costs were 8.266—road was substantially cheaper (Table 1). If these cost differences translated into actual shipping patterns, they would imply a natural sector-mode specialization: agricultural products—bulky, heavy, shipped in large volumes—would fall in the high-density regime where rail has a cost advantage, while manufactured goods—lighter, more varied, shipped in smaller batches—would fall in the low-density regime where road is cheaper.

Whether these cost differences generate meaningful mode-specific effects on regional economic outcomes is ultimately an empirical question. It is not obvious a priori that they do: if road expansion sufficiently compensated for rail closures, or if firms and workers adjusted quickly to the new transport configuration, the sectoral composition of economic activity might not have changed. We treat the existence of mode-specific effects as a hypothesis to be tested in Sections 5 and **??**, rather than as an established fact. The cost data presented here motivate the hypothesis but do not settle it.

3 Data

The empirical analysis combines three data components. Section 3.1 describes the transport network geometries and their changes between 1960 and 1986. Section 3.2 describes the district-level population and economic outcomes. Section 3.3 describes how we construct market access from transport cost surfaces.

3.1 Transport networks

We combine two georeferenced networks: the railway network at its 1979 status, which we map to 1960 and 1986 via documented closure dates, and the paved-and-gravel road network at three cross sections (1954, 1970, 1986).

Railways. The rail network comes from Argentina's 1979 *Política Ferroviaria* report (Fraternidad, La, 1981), Figure V–5, scanned and digitised in QGIS using IGN shapefiles as geographic references. The resulting shapefile contains 565 MULTILINESTRING segments in WGS 84, each carrying three attributes that identify its operating status:

- `status1979` with three values. Value 1 indicates the segment was active in both 1979 and 1986 and is therefore counted in both the 1960 and 1986 rail stocks. Value 2 indicates the segment was closed during the military dictatorship (1976–1983). Value 3 indicates closure between 1960 and 1966; we count these segments in the 1960 stock but not the 1986 stock.

- `studied_co` with two values. Value 1 indicates the segment was one of the 237 rail segments studied in the 1962 Larkin Plan, which recommended closures on the basis of freight-density and operating-cost criteria. Value 0 indicates the remaining 328 segments that the Larkin Plan did not evaluate.

- `recom_code` with three values (1, 2, 3) encoding the Larkin Plan's three operating recommendations for the studied segments.

We intersect the rail segments with the 312 district polygons from the IPUMS time-invariant geography (described in Section 3.2), compute per-district length in kilometres in EPSG:4326 using `sf::st_length` with s2 geometry, and obtain `tot_rails_1960` and `tot_rails_1986` for each district. Total rail km falls from 42,780 in 1960 to 33,303 in 1986, a 22% reduction. Of the 9,477 km lost, 5,563 km (59%) were closed during the dictatorship and 3,914 km (41%) were closed between 1960 and 1966. Figure ??a shows the geography of the rail network and highlights the dictatorship closures in red. 30 of the 312 districts have zero rail km in all periods; these are concentrated in Patagonia and Tierra del Fuego.

Roads. The road network comes from a digitised cross-panel of Automóvil Club Argentino maps for 1954, 1970, and 1986. The resulting shapefile (`comparacion_54_70_86.shp`) contains 1,741 MULTILINESTRING segments classified into seven `type2` categories by their period-specific presence:

- `type2` $\in \{1, 5\}$: present in 1954 and 1986 (type 5 is absent from the 1970 map and treated as a cartographic error).
- `type2` $\in \{2, 3\}$: added between 1954 and 1986 (type 2 appears in 1970, type 3 appears in 1986).
- `type2` $\in \{4, 6, 7\}$: present only in a subset of periods and absent in 1986; these segments are excluded from period-total road km.

The paved-and-gravel road network (the union of types 1, 2, 3, and 5) grows from 27,513 km in 1954 to 79,820 km in 1986, a 190% increase. Figure ??b shows the road network and highlights the segments added between 1954 and 1986 in red.

Three districts, all in Patagonia, lack any segment in the road shapefile and contribute missing values in the district-level road variables. 11 districts lost rail without gaining roads and 141 districts both lost rail and gained roads. The cross-district correlation between Δ(rail km) and Δ(road km) is -0.25.

Hypothetical road networks. We also construct four hypothetical road networks that connect Argentina's largest cities under stylised routing rules. Starting from a node set of 54 cities (all provincial capitals plus all cities with population $\geq$ 15,000 in 1960), we compute:

- `euc`: all $\binom{54}{2} = 1{,}431$ straight-line (Euclidean) segments between city pairs.
- `euc_mst`: the Euclidean minimum spanning tree (53 edges).

- `lcp`: all bilateral least-cost paths between city pairs, routed on a Faber-2014 construction-cost raster that penalises slope, wetland, urban areas, and water crossings.
- `lcp_mst`: the minimum spanning tree of the LCP weights.

These networks serve as instruments for the actual road network in Section 4.

3.2 Census data and outcomes

Geographic unit. We work at the district level using IPUMS International's time-invariant `geolev2` geography for Argentina, covering 312 mainland districts including Capital Federal and the two Tierra del Fuego jurisdictions. The same polygons are used across the five IPUMS sample years (1970, 1980, 1991, 2001, 2010), the 1960 and 1947 census publications, and the agricultural and industrial censuses. The `geolev2` identifier is stored as a character string throughout the analysis to avoid type-coercion bugs on leading zeros.

Population. Population in 1960 comes from the digitised *IV Censo Nacional de Población 1960* tables, reconciled to the IPUMS district geography through a crosswalk that handles pre-1960 splits and name changes. The mean 1960 district population is 43,856 with a national total of 13.6 million. Population in 1991 comes from the IPUMS microdata aggregated to `geolev2`; the mean 1991 district population is 114,723 with a national total of 35.8 million. Log population change $\Delta \log(\text{pop}_{91/60})$ has mean +1.02 and standard deviation 0.53.

Urban population. Urban population shares at the district level come from the 1960 census (classification per INDEC) and from the IPUMS 1991 microdata. Urban population is not coded consistently in the 1970 and 2010 IPUMS microdata for Argentina; those years are recorded as missing in the panel and are not used for urban-outcome regressions. 1960 and 1991 give us the post-reform window with consistent urban/rural classification.

Pre-reform placebo. We supplement the baseline with the 1947 national census, which covers 240 of the 312 districts. A log population change $\Delta \log(\text{pop}_{60/47})$ for this subsample serves as a placebo outcome: it measures population growth before any of the 1960–1986 network changes, and should be uncorrelated with the instruments if the instruments isolate post-1960 network-induced variation.

Industrial and agricultural outcomes. Four additional data components carry sectoral outcomes. The 1954 and 1985 industrial censuses (*Censo Industrial* and *Censo Económico*) provide district-level establishment counts, salaries, and production values. Only `nestab` (establishments), `massal` (wages), and `valprod` (production value) are present in both census years; `nemp` (workers, 1954) and `npers` (persons, 1985) are defined differently and are not directly comparable. The 1960 and 1988 agricultural censuses provide `nexp` (farm operations) and `areatot_ha` (total agricultural area). All monetary variables are nominal pesos of their census year; real comparisons across years require a deflator that we defer to robustness.

Geographic controls. District-level controls come from external sources. Area in km^2 is computed from the IPUMS polygon. Mean elevation comes from GTOPO30 (U.S. Geological Survey, 1996). Terrain ruggedness uses the Nunn–Puga TRI raster (Nunn and Puga, 2012) derived from GTOPO30. Wheat potential yield comes from FAO–GAEZ 3.0 (FAO and IIASA, 2012). Caloric potential (pre- and post-1500) comes from Galor and Özak (2016). Distance to Buenos Aires is the great-circle distance from the centroid of the Gran Buenos Aires polygon in the IGN settlements shapefile. All continuous controls are standardised (zero mean, unit variance) for use in the regressions.

3.3 Market access

Definition. Following Donaldson and Hornbeck (2016), we define district i's market access in period t for sector s under elasticity θ as:

$$\mathrm{MA}_{i,t,s,\theta} = \sum_{j \neq i} \frac{\mathrm{pop}_{j,1960}}{(\tau_{ij,t,s})^{\theta}}, \tag{1}$$

where $\tau_{ij,t,s}$ is the origin–destination transport cost between the centroids of districts i and j under network configuration t and sector s. Population weights are held fixed at their 1960 values across all t so that variation in $\mathrm{MA}_{i,t,s,\theta}$ comes entirely from variation in transport costs rather than from endogenous population responses to the network changes.

Cost surface. Transport costs $\tau_{ij,t,s}$ are the least-cost-path distances on a $2{,}399 \times 3{,}090$ raster covering mainland Argentina in the ESRI:54034 equal-area projection (approximately 1.2 km per cell). The per-pixel cost $c_i(t, s)$ is:

$$c_i(t,s) = \begin{cases} \mathrm{cost}_{\mathrm{nav}}(s) & \text{if pixel } i \text{ is navigable water,} \\ \mathrm{cost}_{\mathrm{mininfra}}(s) & \text{if pixel } i \text{ has both rail and road,} \\ \mathrm{cost}_{\mathrm{rail}}(s) & \text{if pixel } i \text{ has rail only,} \\ \mathrm{cost}_{\mathrm{road}}(s) & \text{if pixel } i \text{ has road only,} \\ \mathrm{cost}_{\mathrm{land}} \cdot \mathrm{HMI}_i & \text{if pixel } i \text{ is off-network land in Argentina,} \\ \mathrm{NA} & \text{otherwise (hard barrier).} \end{cases}$$

Network layers (rail, road, and navigation) are rasterised from the respective shapefiles after a 1 km buffer to guarantee at least one full raster cell per linear segment. The Strait of Magellan is buffered by 5 km so that at least one raster cell per shore becomes navigable, which lets Dijkstra cross the strait and keeps Tierra del Fuego connected to the mainland. Off-network pixels outside

Argentina are set to NA, which acts as a hard barrier for the least-cost path algorithm except where a navigable-water pixel overrides the country mask (e.g. the Paraná river border).

Cost parameters. Sector-specific cost parameters come from Baumgartner and Palazzo (1969) Table II. Three sectors $s \in \{0, 1, 2\}$ correspond to three cargo density scenarios (medium, high, low); Table 1 summarises the values.

Table 1: Transport Costs by Mode and Cargo Density (1960 pesos per ton-km)

	Low density (100 t/day)	Medium density (500 t/day)	High density (1,000 t/day)
Road	8.266	1.777	1.000
Railroad	13.490	1.874	0.465
Navigation	0.621	0.621	0.621

Notes: Ground-transport costs from Baumgartner and Palazzo (1969). Navigation costs from Larkin (1962). Medium-density costs are used in the main analysis (sector $s = 0$); high- and low-density costs (sectors $s = 1$ and $s = 2$) are reported in extensions exploring sector-specific effects.

In particular, rail is cheaper than road at high cargo density ($\text{cost}_{\text{rail}}(s{=}1) = 0.465 < \text{cost}_{\text{road}}(s{=}1) = 1.000$) but more expensive at low density ($\text{cost}_{\text{rail}}(s{=}2) = 13.490 > \text{cost}_{\text{road}}(s{=}2) = 8.266$), which is the scale-economies channel that motivates sector-specific analysis. At medium density, rail and road costs are nearly identical (1.874 vs. 1.777), so mode choice matters less for the overall market-access index. We therefore use the medium-density scenario as the main specification and report the other two as extensions. The off-network cost $\text{cost}_{\text{land}}$ is $17.9 \times \text{cost}_{\text{road}}(s{=}2)/\overline{\text{HMI}}$, where $\overline{\text{HMI}} = 0.2018$ is the national mean of the Özak (2018) Human Mobility Index. The factor 17.9 is the ratio of motor-vehicle speed to walking-equivalent speed, calibrated on a Buenos Aires–Rosario comparison. This gives $\text{cost}_{\text{land}} \approx 733.2$; the per-pixel off-network cost scales with the local HMI value and is one to two orders of magnitude above on-network cost.

Least-cost path and elasticity. We compute $\tau_{ij,t,s}$ by running Dijkstra's algorithm on an eight-connected grid transition object (`gdistance::costDistance`) between the 312 IPUMS centroid points. $\tau_{ii,t,s} = 0$ by construction and τ is symmetric. Dijkstra is run independently for each (period, sector) combination; a single end-to-end run of the four main networks (actual 1960, actual 1986, `stu` instrument, `lcp_mst` instrument) at three sectors takes approximately 45 minutes on a laptop with four cores.

The elasticity θ controls how sharply distant destinations are discounted in the MA sum. We follow the applied-trade literature (Donaldson and Hornbeck 2016; Eaton and Kortum 2002) and report two values, $\theta_{\text{low}} = 4.55$ and $\theta_{\text{high}} = 8.11$. Main results use θ_{low}; θ_{high} appears in the sensitivity tables. [PLACEHOLDER: confirm citations and whether to add a literature-review paragraph here per the `Plan/tasks.md` pending-decision item.]

Change in log MA. The main treatment variable is $\Delta \log \mathrm{MA}_i \equiv \log \mathrm{MA}_{i,1986,s=0,\theta_{\mathrm{low}}} - \log \mathrm{MA}_{i,1960,s=0,\theta_{\mathrm{low}}}$. Across the 312 districts, $\Delta \log \mathrm{MA}_i$ has mean $+1.56$, standard deviation 2.48, minimum -7.18 and maximum $+9.58$; 91% of districts see a positive change. Figure **??** maps $\Delta \log \mathrm{MA}_i$ across districts.

Counterfactual decomposition. We additionally construct two counterfactual cost surfaces that freeze one network mode at its baseline while letting the other change:

- *only-rail*: 1986 rails combined with 1954 roads. $\Delta \log \mathrm{MA}_i^{\text{only-rail}} = \log \mathrm{MA}(\text{only-rail})_i - \log \mathrm{MA}_{i,1960}$.
- *only-road*: 1960 rails combined with 1986 roads. $\Delta \log \mathrm{MA}_i^{\text{only-road}}$ defined analogously.

At sector 0 and θ_{low}, $\Delta \log \mathrm{MA}^{\text{only-rail}}$ has mean -0.40 (rail contraction is a small market-access loss) and $\Delta \log \mathrm{MA}^{\text{only-road}}$ has mean $+1.75$ (road expansion is a large gain). The sum of the two isolated components ($+1.35$) is slightly below the total $\Delta \log \mathrm{MA} = +1.56$, reflecting a positive complementarity between the two network shocks. Figure **??** compares the three maps side by side.

Limitations. The cost-surface construction imposes three modelling choices that the Section 4 discussion of identification returns to. First, mode switches are free: Dijkstra can step from a rail pixel to a road pixel at any grid boundary, whereas in reality such switches occur at stations or terminals and incur transshipment costs. Second, the algorithm selects a single cheapest path, not a probability distribution over routes; in equilibrium firms use multiple parallel routes. Third, cost parameters are constant across space and time; we do not model congestion or regional cost heterogeneity. Each of these simplifications follows the approach in Donaldson and Hornbeck (2016) and is common in the market-access literature.

4 Empirical Strategy

4.1 Estimating Equation

Our main specification relates changes in district-level outcomes to changes in market access:

$$\Delta Y_i = \beta \cdot \Delta \ln \mathrm{MA}_i^{\text{full}} + X_i' \gamma + \varepsilon_i, \tag{2}$$

where ΔY_i is the change in outcome, $\Delta \ln \mathrm{MA}_i^{\text{full}}$ is the change in log market access, and X_i includes baseline log market access, baseline log population, and geographic characteristics. The parameter β is the elasticity of regional outcomes with respect to market access.

4.2 Endogeneity Concerns

4.3 Instrumental Variables

4.3.1 Instrument 1: Larkin Plan Discontinuity

4.3.2 Instrument 2: Hypothetical Road Networks

4.3.3 First Stage

The first stage regression is:

$$\Delta \ln \mathrm{MA}_i^{\text{full}} = \alpha_1 \cdot \Delta \ln \mathrm{MA}_i^{\text{LP}} + \alpha_2 \cdot \Delta \ln \mathrm{MA}_i^{\text{hypo}} + X_i'\delta + u_i. \tag{3}$$

4.4 Validation Package

Sections 4.4.1 and 4.4.2 present two validation exercises that probe the exclusion restriction underlying the two instruments. Section 4.4.3 reports the first stage. A third validation exercise, the over-identification test, is reported alongside Section 5.

4.4.1 Table 6 — Pre-Period Balance

Table **??** reports regressions of each baseline district characteristic on the two instruments, partialling out baseline log market access and baseline log population. Each row is a separate regression of the row-label characteristic; columns report the instrument coefficients under three specifications: the Larkin Plan instrument alone (column 1), the hypothetical-road instrument alone (column 2), and both instruments jointly (column 3). Standard errors are heteroskedasticity-robust.

The Larkin Plan instrument is uncorrelated with baseline log population (column 1 coefficient -0.006, $p = 0.78$) and baseline urban share (0.012, $p = 0.02$, marginal). It shows residual correlations with the six standardized geographic controls: wheat suitability (-0.047, $p < 0.001$), pre-1500 caloric potential (-0.008, $p = 0.004$), post-1500 caloric potential (0.007, $p = 0.003$), distance to Buenos Aires (-0.042, $p < 0.001$), and log district area (-0.144, $p < 0.001$). These correlations are small in magnitude but statistically detectable. Because these same characteristics are controls in the main specification, the residual correlations shift from exclusion-restriction threats to a conditional-independence requirement: the Larkin Plan instrument must be uncorrelated with the outcome *conditional on* these geographic controls.

The hypothetical-road instrument is uncorrelated with all nine baseline characteristics at conventional significance levels: none of the column 2 coefficients has $p < 0.05$. The joint specification in column 3 yields similar results to the single-instrument columns.

4.4.2 Table 7 — Pre-Trends

Table **??** reports a placebo test. The dependent variable is the log change in district population between 1947 and 1960 — a change that predates the 1960–1986 infrastructure restructuring. The regressor is $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ computed over 1960–1986. If the infrastructure changes in 1960–1986 are driven by instruments orthogonal to pre-period district-level dynamics, the coefficient on post-reform $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ in this placebo regression should be near zero.

The sample is 235 districts — the subset for which the 1947 census provides comparable population counts under IPUMS geolev2 boundaries, as discussed in Section 3. Columns (1) to (4) match the structure of Tables **??** and **??**: OLS, IV-LP, IV-Hypo, IV-Both.

The OLS coefficient is 0.035 (0.017), significant at the five-percent level. The IV-LP estimate is 0.070 (0.050), the IV-Hypo estimate is 0.173 (0.319), and the combined IV estimate is 0.078 (0.041), significant at the ten-percent level. The first-stage F for the IV-Hypo column on this sample of 235 districts is 0.91, below conventional weak-instrument thresholds; the IV-Hypo column is therefore not interpretable in this specification. The first-stage F for IV-LP is 17.84 and for IV-Both is 10.21.

The OLS and IV-Both point estimates for the placebo outcome are non-zero and marginally statistically significant. Two readings are consistent with the evidence. First, there may be a genuine pre-existing trend correlated with the eventual infrastructure changes of 1960–1986: districts that were growing fastest in 1947–1960 may have been more likely to gain (or lose) market access in the later period. Second, the placebo outcome is defined on a subsample ($N = 235$) that is non-random with respect to the full estimation sample ($N = 309$), and the correlations documented in Section 4.4.1 for the Larkin Plan instrument may interact with that subsample selection.

Section 5.5 reports the main-spec population regression refit on the same 235-district subsample. On that subsample, the IV-Both coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is 0.067 (0.036), with $p = 0.062$. This coefficient is similar in magnitude to the placebo coefficient of 0.078 reported above. The two readings offered in the preceding paragraph are therefore not distinguishable by this test: the pre-reform pattern on the 235 subset is of the same order as the post-reform main-spec coefficient on the same subset. We retain the main specification on the full sample as the headline estimate and report the placebo-subsample result in the robustness panel.

4.4.3 Table 8 — First-Stage Results

Table **??** reports the first-stage regressions. Results are described in Section 5.1.

4.5 Additional Robustness Approaches

We assess robustness by varying the distance decay parameter (θ), the set of controls, the sample (excluding major cities, border regions), and the time period (1960–1970, 1970–1991). Results are reported in Table **??**.

5 Main Results

5.1 First Stage

Table **??** reports the first-stage regressions corresponding to equation (3). Column (1) uses only the Larkin Plan instrument $\Delta \ln \mathrm{MA}_i^{\mathrm{LP}}$, column (2) uses only the hypothetical-road instrument $\Delta \ln \mathrm{MA}_i^{\mathrm{hypo}}$, and column (3) uses both jointly. All columns include baseline log market access, baseline log population, and the six standardized geographic controls. Standard errors are heteroskedasticity-robust.

The Larkin Plan instrument has a coefficient of 0.240 with a robust standard error of 0.054 in column (1). The corresponding first-stage F statistic is 19.97. The hypothetical-road instrument has a coefficient of 0.123 (0.069) in column (2), with a first-stage F of 3.14. In the joint specification in column (3), both instruments retain predictive power: the Larkin Plan coefficient is 0.254 (0.055) and the hypothetical-road coefficient is 0.146 (0.063); the joint F is 13.05.

The Larkin Plan instrument alone exceeds the Stock–Yogo rule-of-thumb threshold of 10. The hypothetical-road instrument alone does not. The joint specification exceeds the threshold and is adopted as the main specification throughout this section. The hypothetical-road-only column is reported so that the two sources of variation can be inspected separately.

5.2 Population Effects

Table **??** reports estimates of equation (2) for four population outcomes: the log change in total population, in urban population, and in rural population between 1960 and 1991, and the level change in the urban share over the same period. Each outcome panel reports four columns: OLS in column (1), instrumental variables using the Larkin Plan instrument in column (2), the hypothetical-road instrument in column (3), and both instruments in column (4).

For total population (Panel A), the OLS coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is 0.022 (0.012), significant at the ten-percent level. The IV-LP estimate is 0.042 (0.042), the IV-Hypo estimate is 0.059 (0.082), and the combined IV estimate is 0.046 (0.033). The sample is 309 districts.

For urban population (Panel B), the OLS coefficient is 0.030 (0.013), significant at the five-percent level. The IV-LP estimate is 0.062 (0.046), the IV-Hypo estimate is 0.049 (0.129), and the combined IV estimate is 0.060 (0.040). The sample is 284 districts, reflecting districts with positive urban population in both census years.

For rural population (Panel C), the OLS coefficient is 0.031 (0.025). The IV-LP estimate is 0.107 (0.108), the IV-Hypo estimate is 0.119 (0.165), and the combined IV estimate is 0.111 (0.087). The sample is 272 districts.

For the urban share (Panel D), all four estimates are close to zero and statistically indistinguishable from zero (OLS: 0.004 (0.005); IV-Both: 0.002 (0.017)). A one-unit increase in $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ corresponds to a change in the urban share of less than one percentage point.

Across the three log-change outcomes, the IV point estimates exceed the OLS estimates by a factor of approximately two to four. This pattern is consistent with a downward bias in OLS that would arise if infrastructure changes were concentrated in districts with declining pre-period trends; the direction of bias was a possibility raised in Section 4. The standard errors on the IV estimates are large enough that individual point estimates are not statistically distinguishable from zero at conventional levels. The IV-Both column reports the narrowest confidence intervals among the three IV columns.

5.3 Sectoral Activity

Table **??** reports estimates of equation (2) for five sectoral outcomes drawn from the industrial and agricultural censuses. Panel A covers three manufacturing outcomes from the 1954 and 1985 industrial censuses: the log change in the number of establishments, in the value of production, and in the total wage mass. Panel B covers two agricultural outcomes from the 1960 and 1988 agricultural censuses: the log change in the number of farms and in the total farmed area. Column structure matches Table **??**.

For manufacturing establishments, the OLS estimate is 0.025 (0.017) and the combined IV estimate is 0.055 (0.061). Neither is statistically distinguishable from zero.

For manufacturing production value, the OLS estimate is 0.060 (0.037). The IV-LP estimate is 0.239 (0.138), significant at the ten-percent level. The combined IV estimate is 0.259 (0.119), significant at the five-percent level. The sample is 308 districts.

For manufacturing wage mass, the OLS estimate is 0.092 (0.035), significant at the one-percent level. The IV-LP estimate is 0.258 (0.146), significant at the ten-percent level. The combined IV estimate is 0.330 (0.126), significant at the one-percent level. The sample is 307 districts.

For the number of agricultural farms (Panel B), the OLS estimate is −0.011 (0.012). The combined IV estimate is 0.095 (0.079). Neither is statistically distinguishable from zero.

For total agricultural farmed area, the OLS estimate is −0.001 (0.018). The combined IV estimate is 0.104 (0.078). Neither is statistically distinguishable from zero.

The estimates differ substantially across the two panels. The manufacturing outcomes related to activity intensity — production value and wage mass — respond strongly and positively to $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$: combined-IV point estimates are 0.259 and 0.330, respectively. The number of manufacturing establishments does not respond detectably. The agricultural outcomes show point estimates close to zero with wide confidence intervals. Section 7 returns to the question of whether this pattern is consistent with the scale-economies hypothesis introduced in Section 2.

5.4 Other Outcomes

Table **??** reports estimates of equation (2) for four outcomes drawn from IPUMS microdata: the level change in the share of adults with college education, the level change in the share with secondary education, the level change in the share of residents who migrated in the past five years, and the level change in the employment rate (the share of individuals classified as employed in IPUMS EMPSTAT). All four outcomes are level differences in population shares between 1970 and 1991.

The regressor remains $\Delta \ln \mathrm{MA}_i^{\text{full}}$ computed over 1960–1986; infrastructure restructuring predates the 1970 outcome measurement. Column structure matches Table **??**.

For the college share, all four estimates are statistically indistinguishable from zero, with point estimates between -0.003 and 0.001. For the secondary share, the OLS and combined-IV point estimates are 0.000; the IV-LP estimate alone is 0.005 ($p = 0.04$).

For the recent-migration share, the OLS estimate is -0.007 (0.002), significant at the one-percent level. The combined IV estimate is -0.020 (0.009), significant at the five-percent level. The sample is 309 districts. A one-unit increase in $\Delta \ln \mathrm{MA}_i^{\text{full}}$ corresponds to a 0.020 decrease in the share of residents reporting recent migration.

For the employment rate, the OLS estimate is -0.001 (0.002) and the combined IV estimate is -0.010 (0.005), marginally significant at the ten-percent level.

The direction of the recent-migration coefficient runs opposite to a naive prediction that market-access gains attract in-migrants. Two readings are consistent with the sign. First, high-market-access districts may have retained their existing populations, producing lower recent-migrant shares because turnover was low. Second, districts that lost population through out-migration to high-market-access receiving districts may have ended with small remaining populations among whom a given absolute inflow of recent migrants generates a higher share. Distinguishing these readings requires data on migration flows by origin and destination, which are beyond the cross-sectional sample available here.

The window for these four outcomes is 1970–1991 because Argentine IPUMS microdata begins in 1970, one decade after the start of the estimation window for Table **??**. The infrastructure restructuring and the bulk of rail closures occurred before 1970, so the timing of outcome measurement post-dates treatment.

5.5 Robustness

Table **??** reports three stress tests of the main population elasticity estimated in Section 5.2 (Panel A of Table **??**): the coefficient on $\Delta \ln \mathrm{MA}_i^{\text{full}}$ is 0.046 (0.033) under the combined IV specification on the full 309-district sample.

Panel A — alternative trade elasticity. The main specification sets the trade elasticity $\theta = 4.55$. Panel A reports the same four specifications with $\theta = 8.11$, which corresponds to a higher-elasticity calibration in the market-access literature. All MA variables, including the treatment, the two instruments, and the baseline log-MA control, switch to the $\theta = 8.11$ variant. The combined IV coefficient is 0.020 (0.017). The coefficient is approximately half the main-specification value. The direction of the effect is unchanged, and no coefficient flips sign.

Panel B — alternative hypothetical-road instruments. The main specification uses the least-cost-path minimum spanning tree as the hypothetical-road instrument. Panel B reports results using three alternative hypothetical networks: the Euclidean minimum spanning tree, the full bilateral least-cost path network, and the full bilateral Euclidean network. Holding θ at 4.55, the combined-IV coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ ranges from 0.019 to 0.062 across the four choices. Individual IV-Hypo columns show wide variation; the large point estimates and standard errors in that column reflect weak-instrument noise in some of the alternative networks and are not interpretable in isolation. The combined-IV column is more stable.

Panel C — sample robustness. Panel C refits the main specification on the 235-district subsample for which the 1947 placebo outcome is defined, as reported in Section 4.4.2. On this subsample, the OLS coefficient is 0.027 (0.013), significant at the five-percent level, and the combined IV coefficient is 0.067 (0.036), significant at the ten-percent level. Both estimates are larger than their full-sample counterparts.

The Panel C estimate bears on the interpretation of Section 4.4.2. The placebo coefficient on the 1947–1960 population change was 0.078 on the same 235-district subsample. The main-specification coefficient on the 1960–1991 population change on the same subsample is 0.067. The pre-reform coefficient and the post-reform coefficient are of the same order of magnitude on the common sample. This co-movement is consistent with both candidate readings offered in Section 4.4.2 and does not adjudicate between them.

6 Mode-Specific Patterns via Counterfactuals

[PLACEHOLDER: Section 6 pending (Block 2). Counterfactual MA measures (rail-only, road-only) not yet computed; deferred until Block 1 is stable.**]**

7 Mechanisms and Heterogeneity

[PLACEHOLDER: Section 7 pending (Block 2). Local-infrastructure regressions (stations, depots) and heterogeneity analysis.**]**

8 Conclusion

[PLACEHOLDER: Section 8 pending – write last.**]**

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